

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

UC LINA 600 Linear Algebra for Graphics — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Strang — Introduction to Linear Algebra, with MIT 18.06 (OCW, full video)	The primary course, free in full. The lectures are the standard introduction for good reason, and the column-space framing is exactly the one this course uses.
3Blue1Brown — Essence of Linear Algebra (free video)	Watch this first, all of it, before anything else. Fifteen short videos that build the geometric intuition the algebra then formalises — and Modules 03, 04, and 09 are much easier afterwards.
Immersive Math — Interactive Linear Algebra (free)	Modules 02 through 06, free. Every figure is manipulable, which matters more here than in most subjects.
Lengyel — Mathematics for 3D Game Programming and Computer Graphics	Modules 06 through 11. The graphics-specific treatment, including quaternions done properly. Library copy.
Gortler — Foundations of 3D Computer Graphics	Modules 04, 07, and 08. Unusually careful about frames and change of basis, which is exactly where graphics texts are usually sloppy. Library copy.
Shirley — Ray Tracing in One Weekend (free)	Project 2's companion. Free, short, and it exercises Modules 05, 06, and 09 on a real renderer.

Module-by-module map

Module 01 — Why This Is the Graphics Prerequisite

Resource	What it covers for this module
3Blue1Brown — Essence of Linear Algebra (free)	Watch all fifteen before Module 02 — it is three hours and it changes how every later module reads.
Scratchapixel — the geometry and transform sections (free)	§1's pipeline , derived step by step with code, and free.
MIT 18.06, lecture 1 (free video)	§2 — Strang's opening, which frames the whole subject around the column picture.
Gortler, chapters 2 and 3 (library copy)	§§1 and 2 — frames and transforms stated more carefully than most graphics books manage.

Module 02 — Vectors, Spans, and Bases

Resource	What it covers for this module
3Blue1Brown, chapters 1 to 3 (free)	The whole module — vectors, span, and basis, visually, and this is the right first exposure.
MIT 18.06, lectures 1 and 5 (free video)	§§2 and 3 — the column picture and independence, from Strang.
Immersive Math, chapters 2 and 3 (free)	§§1 to 3 with manipulable figures — drag the basis vectors and watch the coordinates change, which is the point of §1.
Gortler, chapter 2 (library copy)	§4's frames — unusually careful about labelling which space a vector is in, which most graphics texts are not.

Module 03 — Matrices as Linear Maps

Resource	What it covers for this module
3Blue1Brown, chapters 3 to 5 (free)	§§1 and 2 — matrices as transformations and multiplication as composition, which is this module's entire thesis.
MIT 18.06, lectures 2 and 3 (free video)	§§1 and 4 — the column picture, which Strang builds the course on.
Immersive Math, chapter 6 (free)	§§1 to 3 — drag a transform's basis images and watch the matrix change, which is §1 made interactive.
Lengyel, chapters 2 and 3 (library copy)	§3 — the graphics transforms in the conventions you will meet in an engine.

Module 04 — Change of Basis

Resource	What it covers for this module
3Blue1Brown, chapter 13 (free)	§§1 and 2 — change of basis, and the whose-coordinates question made visual.

Resource	What it covers for this module
Gortler, chapters 3 and 4 (library copy)	The whole module — and the frame-naming discipline of §2 comes from here, which is why this book is recommended over the more popular ones.
MIT 18.06, the change of basis lecture (free video)	§2 — the algebra, carefully.
Scratchapixel, the camera and lookAt articles (free)	§3 — the view matrix derived rather than presented, with code.

Module 05 — Dot Products and Projection

Resource	What it covers for this module
3Blue1Brown, chapter 9 (free)	§1 — the dot product's two definitions and why they agree, made visual.
MIT 18.06, the orthogonality lectures (free video)	§§2 to 4 — projection, orthonormal bases, and Gram-Schmidt, with the least-squares connection previewed (Module 12).
Immersive Math, chapter 4 (free)	§§1 and 2 — drag the vectors and watch the projection follow.
Pharr, Jakob & Humphreys — PBRT, the sampling chapters (free online)	§4's graphics use — orthonormal frames around a normal, which is CSCE 647's version of this material.

Module 06 — Cross Products and Normals

Resource	What it covers for this module
3Blue1Brown, chapters 10 and 11 (free)	§1 — the cross product, including the determinant connection that Module 09 develops.
Lengyel, chapters 1 and 4 (library copy)	§§2 to 4 — normals, winding, and the normal matrix, in graphics conventions.
Scratchapixel, the normals and shading articles (free)	§§2 and 4 — with the inverse transpose derived rather than asserted.
The OpenGL and DirectX coordinate system documentation (free)	§3 — the two conventions stated by their own specifications, which is the authoritative source when they disagree.

Module 07 — Homogeneous Coordinates

Resource	What it covers for this module
Gortler, chapters 3 and 11 (library copy)	§§1 to 3 — points versus directions treated properly, which most texts blur.
Scratchapixel, the homogeneous coordinates article (free)	§§1 and 4 — free, with the perspective divide derived step by step.
Lengyel, chapter 4 (library copy)	§3 — the 4×4 structure in engine conventions.

Resource	What it covers for this module
Hartley & Zisserman, chapter 2 (library copy)	§4 rigorously — projective geometry properly, which is what CSCE 753 will want.

Module 08 — The Pipeline as a Matrix Chain

Resource	What it covers for this module
Scratchapixel, the perspective projection series (free)	§§2 and 3 — the derivation done slowly and in full, free, with code.
Gortler, chapters 10 and 11 (library copy)	§§1 and 2 — the pipeline with every frame named, which is the careful version.
Song Ho Ahn — OpenGL projection matrix (free)	§2 — the standard derivation with every algebraic step shown, which is useful for checking yours.
Reversed-Z and depth precision writeups (free)	§3 — depth precision visualised, and the reversed-Z technique that largely solves it.

Module 09 — Determinants and Degeneracy

Resource	What it covers for this module
3Blue1Brown, chapters 6 and 7 (free)	§§1 to 3 — the determinant as a volume factor, which is the framing this module uses throughout.
MIT 18.06, the determinant lectures (free video)	§§1 and 2 — the properties, and the connection to invertibility.
Möller & Trumbore — Fast ray-triangle intersection (free)	§4's fourth row — determinants doing real work in a renderer's inner loop.
Trefethen & Bau — Numerical Linear Algebra	The closing callout — condition numbers and why determinants are the wrong tool for conditioning. Library copy.

Module 10 — Eigenvalues and Eigenvectors

Resource	What it covers for this module
3Blue1Brown, chapters 14 and 15 (free)	§§1 and 3 — eigenvectors geometrically, and the change-of-basis reading of diagonalisation.
MIT 18.06, the eigenvalue lectures (free video)	§§2 to 4 — including the spectral theorem and the applications to difference equations.
Trefethen & Bau, lectures 24 to 29	§2's warning — how eigenvalues are actually computed, and why not by the polynomial. Library copy.

Resource	What it covers for this module
Shlens — A Tutorial on Principal Component Analysis (free)	§4's last point — PCA derived as an eigendecomposition, clearly, and free.

Module 11 — Quaternions

Resource	What it covers for this module
3Blue1Brown and Ben Eater — Visualizing quaternions (free, interactive)	§2 — the best explanation available, and it is interactive, which this topic needs more than most.
Shoemake — Animating rotation with quaternion curves	§3's slerp , in the original paper that introduced it to graphics.
Lengyel, chapter 4 (library copy)	§§2 to 4 — quaternions in engine conventions, with the conversions worked.
Hanson — Visualizing Quaternions	§2 in depth — far more than this course needs, and the right book if you want the full picture. Library copy.

Module 12 — Least Squares

Resource	What it covers for this module
MIT 18.06, the least squares lectures (free video)	§§1 and 2 — projection and the normal equations, with the column-space picture that makes it obvious.
3Blue1Brown, the projection material (free)	§1 — the geometric picture of projecting onto a subspace.
Trefethen & Bau, lectures 11 and 18 to 20	§3 — why QR rather than the normal equations, with the conditioning analysis. Library copy.
Hastie, Tibshirani & Friedman — Elements of Statistical Learning (free PDF)	§4's first row — regression as least squares, free, and the direct bridge to CSCE 633.

Module 13 — Claiming a Transform Is Correct

Resource	What it covers for this module
Gortler, the conventions appendix (library copy)	§4 — conventions stated explicitly, which is what this module is asking you to do.
Scratchapixel, the testing and debugging notes (free)	§2 — practical verification of a renderer, free.
Shirley — Ray Tracing in One Weekend (free)	§§1 and 2 — a complete small renderer to test your understanding against, free and short.
The glTF specification, the coordinate conventions (free)	§4 — a format that states its conventions explicitly, which is why it interoperates.